

09/23/2026: Relations

CSCI 246: Discrete Structures

Textbook reference: Sec 14, Scheinerman

Today's Agenda

- Review group exercises (≈ 10 mins)
- Mini-lecture (≈ 20 mins)
- Group exercises (≈ 15 mins)

Study Guide For Quiz 4

Readings. Sec. 10 (Sets)

- Know how to apply the strategy for proving that two sets are equal (Proof Template 5). In particular, know how to prove Prop 10.1.
- Know how to count the number of subsets of a set.

Sec. 11 (Quantifiers)

- Know how to apply the strategy for proving existential statements (Proof Template 7) to Example 11.1. Know how to apply the strategy for proving universal statements (Proof Template 8) to Example 11.2.
- Know how to read and negate quantified statements.

Sec. 12 (Set Operations)

- From Theorem 12.3, know how to prove the associative properties (see text) and distributive properties (see slides).
- Know how to apply Prop 12.4 (the size of a union) to solve Example 12.5.
- Know how to prove Prop 12.11 (alternative form of symmetric difference).

Group exercises. Know how to do all of the group exercises from the course meetings on Sets (9/16), Quantifiers (9/18), and Set Operations (9/21).

Outline for Mini-Lecture:

- **Introduction**
- Drawing relations as pictures
- Properties of relations
- Drawing properties as pictures
- Applications

Relations

A **relation** is just a set of ordered pairs that connect elements of one set to another (or the same set).

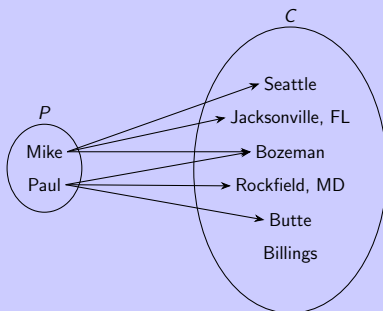
Relations

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Example

- Let P be a set of people. Let C be a set of cities.
- Define the relation $R \subseteq P \times C$ by

$$R = \{(p, c) : \text{person } p \text{ has lived in city } c.\}$$



Example

Let $A = \{2, 3, 4, 5, 6\}$ and $B = \{3, 6, 9\}$ and relation $R = \{(a, b) : a|b, a \in A, b \in B\}$. Then $R =$

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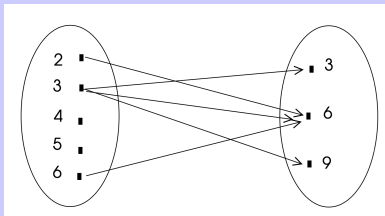


Figure 1: Set diagram representation of the relation R

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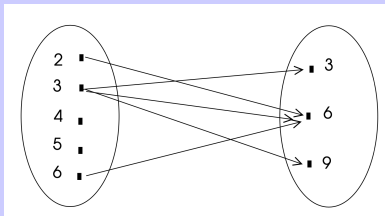


Figure 1: Set diagram representation of the relation R

Definition (Hampkins)

Let A and B be two sets. A **binary relation** (or simply a **relation**) R from A to B is a subset of the Cartesian product $A \times B$.

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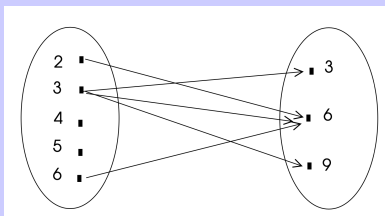


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Let A and B be two sets. A **binary relation** (or simply a **relation**) R from A to B is a subset of the Cartesian product $A \times B$.

Definition (Scheinerman Def 14.1)

A **relation** is a set of ordered pairs.

Relations: Some odds and ends

Notation

If $(a, b) \in R$, we write aRb .

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Example.

- Let $A = B = \mathbb{Z}$, and $R = "\leq"$.
- Then $2R2$ and $2R3$ but $3 \not R 2$.

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Poll. How is a relation different from a function?

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Poll. How is a relation different from a function?

Solution. An $a \in A$ can connect to **more than one** $b \in B$. For example:

- Let R be the relation has-lived-in-city.
- Let $m = \text{Mike}$, $b = \text{Bozeman}$, $s = \text{Seattle}$.
- Then we have mRb **and** mRs .

Relations: Some odds and ends

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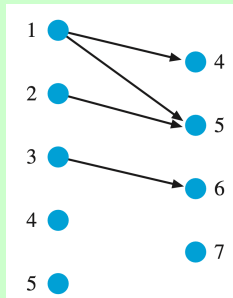
- Introduction
- Drawing relations as pictures
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Drawing relations from A to B

To depict a relation R from A to B , place dots for A on the left and for B on the right, then draw an arrow from $a \in A$ to $b \in B$ whenever $(a, b) \in R$.

Example

- Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{4, 5, 6, 7\}$.
- Let R be the relation $\{(1, 4), (1, 5), (2, 5), (3, 6)\}$.
- Then the figure below shows R as a picture.



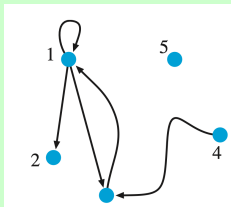
Drawing relations on A

To draw a relation R on A , either:

1. View R as a relation from A to itself and use the previous method.
2. Represent each element of A as a dot and draw arrows for pairs $(a, b) \in R$ (including loops if aRa).

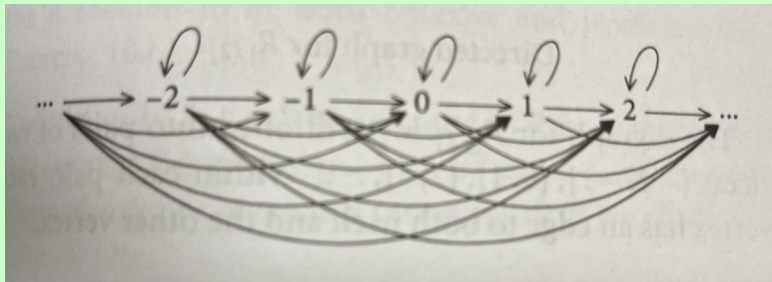
Example

- Let $A = \{1, 2, 3, 4, 5\}$.
- Let $R = \{(1, 1), (1, 2), (1, 3), (4, 3), (3, 1)\}$.
- A picture of the relation R (using option #2) is given below.



An Example Using a Set with Infinitely Many Elements

Let $A = \mathbb{Z}$ and R be the $\leq$ relation. A picture of the relation is given below



Outline for Mini-Lecture:

- Introduction
- Drawing relations as pictures
- **Properties of relations**
- Drawing properties as pictures
- Applications

Definition

Suppose that R is a relation on a set A .

1. The relation R is **reflexive** on A if $a R a$ for all $a \in A$.
2. The relation R is **symmetric** on A if, whenever $a R b$, then also $b R a$.
3. The relation R is **transitive** on A if, whenever $a R b$ and $b R c$, then also $a R c$.

Poll. Let A be the set of people. Can you think of an example and counterexample for each property?

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Let A be the set of people.

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Examples

Let A be the set of people.

1. **Reflexivity.** The lives-in-the-same-city-as relation is reflexive, but the parent-of relation is not.
2. **Symmetry.**

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2. **Symmetry.** The friends-of relation is symmetric (usually – 😊), but parents-of relation is not.
3. **Transitivity.**

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Let A be the set of people.

1. **Reflexivity.** The lives-in-the-same-city-as relation is reflexive, but the parent-of relation is not.
2. **Symmetry.** The friends-of relation is symmetric (usually – 😊), but parents-of relation is not.
3. **Transitivity.** The is-in-the-same-family relation is transitive, but the parent-of relation is not.

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Reflexivity

- A relation is **reflexive** if every element has a self-loop.

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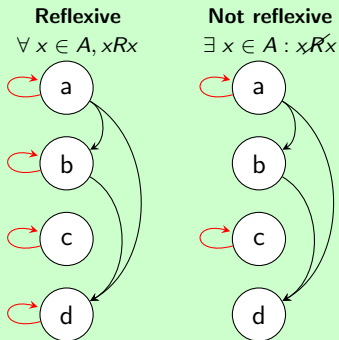
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When one or more red connection drops out, the relation loses reflexivity.

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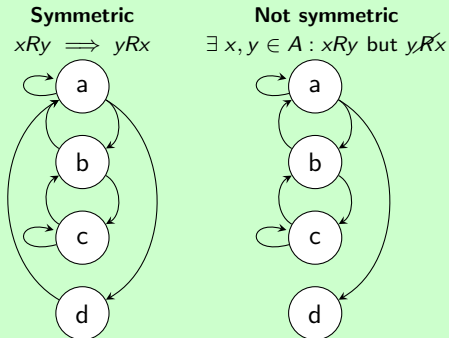
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When we dropped the connection from d to a , the relationship lost symmetry.

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- A relation is **transitive** if every two-hop trip can also be done in a single-hop.

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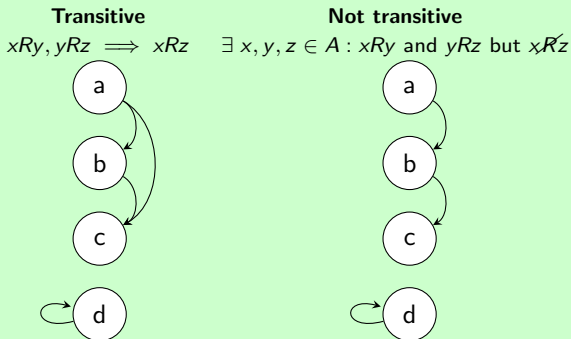
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Example



When we dropped the connection from *a* to *c*, the relationship lost transitivity.

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Example of Relations in Computer Science

Example

Let's consider a "friendship" relation in a **social network** like Facebook. We can define a relation R on a set of users where: $(x, y) \in R$ if and only if x is a friend of y .

Poll

Is this relation

- Reflexive?

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- Symmetric?

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- Reflexive? **✗** Users don't usually friend themselves in social networks.
- Symmetric? **✓** (-ish.) In most social networks, a friendship connection is bidirectional. If Alice is a friend of Bob, then Bob is a friend of Alice. However, in platforms like X (formerly Twitter), where following is *one-way*, the "follows" relation is *not symmetric*.
- Transitive?

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- Reflexive? **✗** Users don't usually friend themselves in social networks.
- Symmetric? **✓** (-ish.) In most social networks, a friendship connection is bidirectional. If Alice is a friend of Bob, then Bob is a friend of Alice. However, in platforms like X (formerly Twitter), where following is *one-way*, the "follows" relation is *not symmetric*.
- Transitive? **✗** If Alice is friends with Bob and Bob is friends with Charlie, it does not necessarily mean that Alice is friends with Charlie.

Why do relations matter?: Some examples from social networks

- **Social Influence & Virality**

- If relationships in a network are **transitive**, messages, trends, and viral content can spread faster.

- **Friend-of-a-Friend Recommendations**

- Even if friendship is not strictly **transitive**, many recommendation algorithms use approximate transitivity to suggest new friends (e.g., LinkedIn's "People You May Know" feature).

- **Algorithm Optimization**

- If a relation is **symmetric**, storage can be optimized by only storing one direction of the connection.
- Search algorithms (e.g., shortest path) can be optimized by reducing redundant checks.

Group exercises

Random Group Assignments

Ashley, Isabelle: 7

Bailey, Tanna: 4

Blaisdell, Olyver: 12

Brown, Ashton: 11

Brown, Jonathan: 10

Cardenas, Dominic: 8

Cutler, George: 1

Fabik, Roland: 3

Fellows, Audrey: 9

Fitzgerald, Ryan: 3

Fuller, Ryan: 10

Griffen, Clara: 5

Grutsch, Tanner: 12

Hager, Nathan: 5

Horner, Jayden: 5

Johnson, Jaret: 2

Kintzel, Samantha: 9

Lameres, Alexis: 7

Lee, Holden: 10

Mayala, Reece: 8

McLean, Connor: 4

Mcbride, Daniel: 2

Miller, Maverick: 11

Moss, Emily: 8

Mousseau, Matthew: 2

Neuman, Evan: 1

Putnam, John: 4

Ramos, Vydel: 9

Rathbun, Jedidiah: 3

Roberts, Matthew: 6

Roller, Nathan: 11

Ryan, Otis: 6

Shevtsov, Timothy: 13

Sinclair, Aidan: 1

Smith, Charles: 13

Stensland, Emma: 7

Thiessen, Logan: 12

Thompson, Jaiden: 6

Vath, Ethan: 13

Group Exercises: Relations

1. (*Relations as ordered pairs.*) Write the following relations on the set $\{1, 2, 3, 4, 5\}$ as sets of ordered pairs: (a) is-less-than, (b) is-divisible-by, (c) is-equal-to, (d) has-the-same-parity-as. (Note: When we say two numbers have the same parity, we mean they are both odd or both even.)
2. (*Drawing pictures of relations.*) Draw pictures of the following relations on A .
 - a. Let $A = \{a \in \mathbb{N} : a|10\}$ and let R be the relation $|$ (divides) restricted to A .
 - b. Let $A = \{1, 2, 3, 4, 5\}$ and let R be the relation $=$ (equals) restricted to A .
3. (*Drawing pictures of relations.*) Draw pictures of the following relations on $A \times B$.
 - a. Let $A = \{1, 2, 3, 4\}$ and $B = \{2, 3, 4\}$. Let R be the relation $\geq$ (greater than or equal to) from A to B .
4. (*Properties of relations.*) For each of the following relations on the set of all human beings, determine whether the relation is reflexive, symmetric, and/or transitive: (a) is-taller-than, (b) has-the-same-parents-as (i.e., same mother and father), (c) is-the-child-of, (d) is-married-to.
5. (*Properties of relations.*) For each of the following relations defined on the integers, please state whether the relation is reflexive, symmetric, and/or transitive: (a) $=$ (equality), (b) $\leq$ (less than or equal to), (c) $<$ (strict less than), (d) $|$ (divides). Please justify your answer for part (d).

Solution to Group Exercise #1

Problem. (*Relations as ordered pairs.*) Write the following relations on the set $\{1, 2, 3, 4, 5\}$ as sets of ordered pairs: (a) is-less-than, (b) is-divisible-by, (c) is-equal-to, (d) has-the-same-parity-as. (Note: When we say two numbers have the same parity, we mean they are both odd or both even.)

Solution.

a.

$$R = \left\{ \begin{array}{l} (1, 2), (1, 3), (1, 4), (1, 5), \\ (2, 3), (2, 4), (2, 5), \\ (3, 4), (3, 5), (4, 5) \end{array} \right\}$$

b.

$$R = \left\{ (1, 1), (2, 1), (3, 1), (4, 1), (5, 1), (2, 2), (4, 2), (3, 3), (4, 4), (5, 5) \right\}$$

c.

$$R = \left\{ (1, 1), (2, 2), (3, 3), (4, 4), (5, 5) \right\}$$

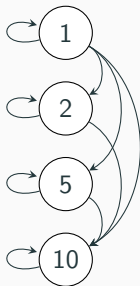
d.

$$R = \left\{ \begin{array}{l} (1, 1), (1, 3), (1, 5), (3, 1), (3, 3), (3, 5), (5, 1), (5, 3), (5, 5), \\ (2, 2), (2, 4), (4, 4), (4, 2) \end{array} \right\}$$

Solution to Group Exercise #2a

Problem. (*Drawing pictures of relations.*) Draw pictures of the following relation on A : Let $A = \{a \in \mathbb{N} : a|10\}$ and let R be the relation $|$ (divides) restricted to A .

Solution. First note that $A = \{1, 2, 5, 10\}$. Now we draw



Solution to Group Exercise #2b

Problem. (*Drawing pictures of relations.*) Draw pictures of the following relation on A : Let $A = \{1, 2, 3, 4, 5\}$ and let R be the relation $=$ (equals) restricted to A .

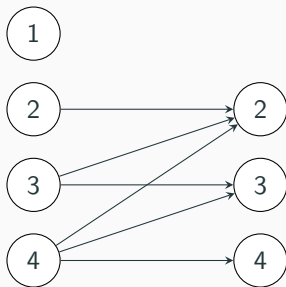
Solution. We draw



Solution to Group Exercise #3

Problem. (*Drawing pictures of relations.*) Draw pictures of the following relations on $A \times B$: Let $A = \{1, 2, 3, 4\}$ and $B = \{2, 3, 4\}$. Let R be the relation $\geq$ (greater than or equal to) from A to B .

Solution. We draw



Solution to Group Exercise #4

Problem. (*Properties of relations.*) For each of the following relations on the set of all human beings, determine whether the relation is reflexive, symmetric, and/or transitive: (a) is-taller-than, (b) has-the-same-parents-as (i.e., same mother and father), (c) is-the-child-of, (d) is-married-to.

Solution.

- a. Transitive.
- b. Reflexive, symmetric, transitive.
- c. None.
- d. Symmetric, transitive (vacuously).

Solution to Group Exercise #5

Problem. (*Properties of relations.*) For each of the following relations defined on the integers, please state whether the relation is reflexive, symmetric, and/or transitive: (a) $=$ (equality), (b) $\leq$ (less than or equal to), (c) $<$ (strict less than), (d) $|$ (divides). Please justify your answer for part (d).

Solution.

1. Reflexive, symmetric, transitive.
2. Reflexive, transitive.
3. Transitive.
4.
 - Reflexive? $\checkmark$ Every number divides itself: $a|a$, since $a \cdot 1 = a$.
 - Symmetric? $\times$ For example, $3|6$ but it is not the case that $6|3$.
 - Transitive? $\checkmark$ Suppose $a|b$ and $b|c$. We need to show $a|c$. Since $a|b$, there is an integer q such that $b = aq$. By definition of $b|c$, there is an integer r such that $c = br$. Hence we have

$$c = br = (aq)r = a(qr)$$

Hence there is an integer $s = qr$ such that $c = as$. Hence $a|c$.